
ASP3231 Project

Lab 4 Feedback

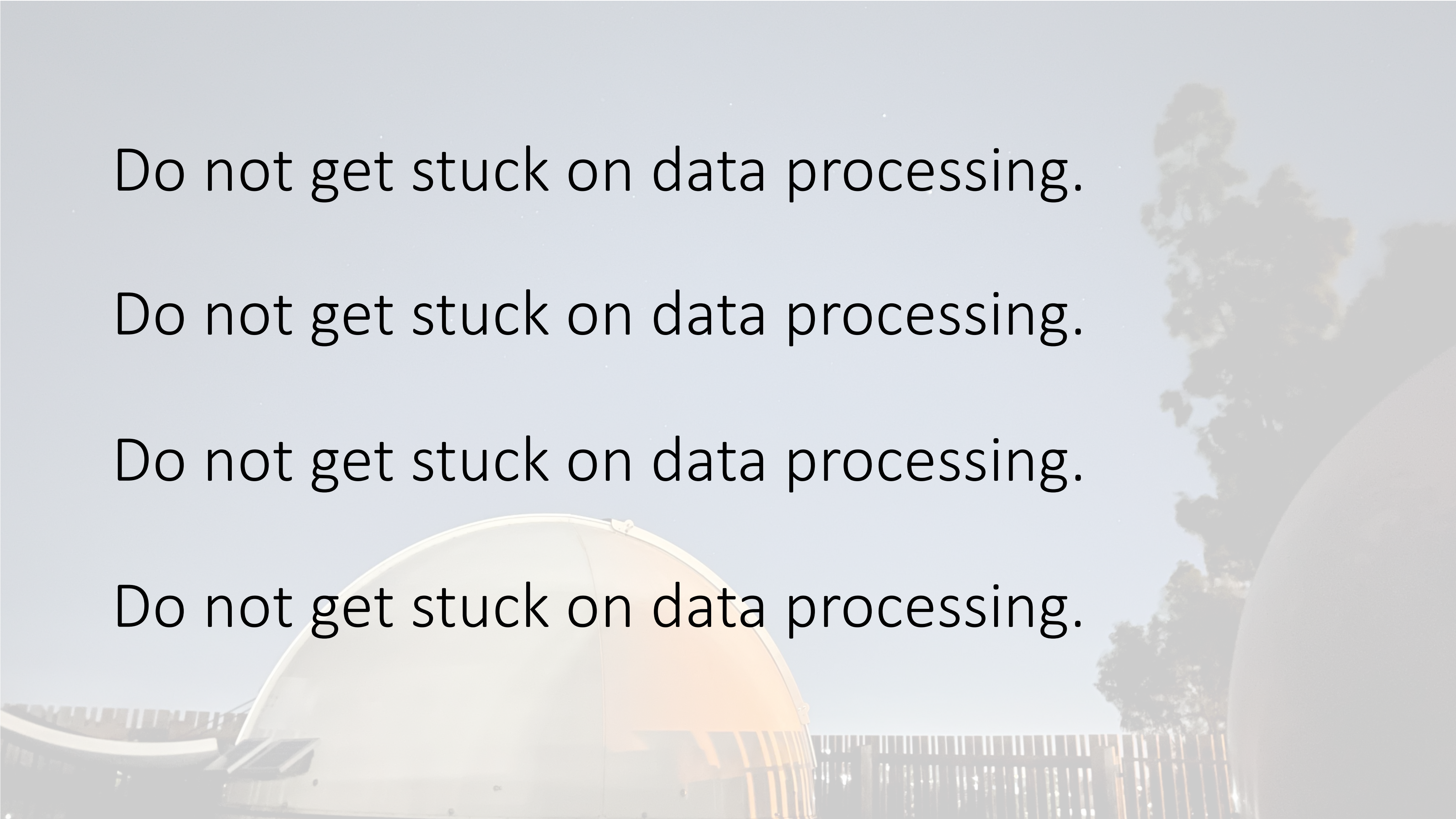
- Prelab done well mostly but I need to update questions
- Image appearance (black-white-grey are set by colour scale).
- Segmentation images
- Photometry – what is more accurate – Gaia or Simbad?
  - Is a V-band magnitude derived from Gaia better than a quoted Simbad V-band mag?
  - Brighter / fainter magnitude is better than higher / lower magnitude
- Note the aperture radius was large – for the project use something smaller
- Consistency checks – numbers and plots
  - Do you know what the numbers mean? Do they matter?
- Comment and discuss your code

Projects

- Collect, process, analyse, interpret and write up observations of an astronomical object.
- Notebook(s) and report. Unassessed sparkler talk.
- Determine distances using the HR diagram (OC, GC, LMC)
- Determine ages using the HR diagram (OC, GC, LMC)
- Determine ages using integrated colours (GC)
- Determine mass-to-light using colours (GC)
- Determine initial mass function using main sequence star luminosities (OC)

Project Data Processing Tips

- 1/3 Data Processing, 1/3 Data Analysis, 1/3 Write Up.
- **Do not get stuck on data processing**
- Do not just use the project sessions on Fridays (Easter and ANZAC day!)
- For SeeStar telescopes you can use data from different telescopes
- Memory limits – one band at a time if needed
- Memory limits – use garbage collection
- Memory limits – could store data as unsigned integers
    - Background subtract but so the background is 1000 (not zero) so all the numbers are positive.
- Shift all science images to common frame (x-y-rotation)
- Scale images using a different reference image for each band
- Your notebook(s) are your log (don’t overwrite cells)
- Multiple notebooks are fine



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Intermediate steps

- Sparkler Talks
- 1 min on your project
- Notebook submissions

Data access

- Automated logs are available for all nights
- The automated logs do not cover image quality etcetera
- You can search the logs for all the data for your object
- SeeStar telescopes sometimes use Caldwell designations (C 106)
- Can look at the images on your device if you cannot remember your object designation and we also have some of that information written into the observing schedule.